



Lab Task – 7

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Course: Programming Fundamentals – Lab

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Task – 1

```
#include<iostream>

using namespace std;

int max_hours ( int arr [ ] )
{
    int max = arr [ 0 ];
    int max_ind = 0;

    for ( int ind = 1; ind < 10; ind++ )
    {
        if ( arr [ ind ] > max )
        {
            max = arr [ ind ];
            max_ind = ind + 1;
        }
    }
}
```

```
    return max_ind;
}

int main()
{
    int student_hours [ 10 ];
    int most_hours = 0;

    cout << "Enter the number of hours studied by 10
students: " << endl;

    for ( int ind = 0; ind < 10; ind++ )
    {
        cin >> student_hours [ ind ];
    }

    most_hours = max_hours ( student_hours );
    cout << endl;
```

```
for ( int ind = 0; ind < 10; ind++ )
{
    if ( student_hours [ ind ] > 1 )
    {
        cout << "Student " << ind + 1 << " studied: "
<< student_hours [ ind ] << " hours." << endl;
    }
    else if ( student_hours [ ind ] == 1 )
    {
        cout << "Student " << ind + 1 << " studied: "
<< student_hours [ ind ] << " hour." << endl;
    }
    else
    {
        cout << "Student " << ind + 1 << " entered
invalid studied hours !" << endl;
    }
}
```

```
}
```

```
cout << endl;
```

```
cout << "Student " << most_hours << " studied  
the most." << endl;
```

```
return 0;
```

```
}
```

Output

```
PS E:\Lab Tasks\Lab Task 7> g++ lab-task7.1.cpp -o lab-task7.1.exe
PS E:\Lab Tasks\Lab Task 7> ./lab-task7.1.exe
Enter the number of hours studied by 10 students:
2
1
3
4
0
6
2
3
8
11

Student 1 studied: 2 hours.
Student 2 studied: 1 hour.
Student 3 studied: 3 hours.
Student 4 studied: 4 hours.
Student 5 entered invalid studied hours !
Student 6 studied: 6 hours.
Student 7 studied: 2 hours.
Student 8 studied: 3 hours.
Student 9 studied: 8 hours.
Student 10 studied: 11 hours.

Student 10 studied the most.
```

Task – 2

```
#include<iostream>

using namespace std;

int main()
{
    int arr [ 7 ];
    int multiple = 0;
    cout << "Enter seven integers: " << endl;

    for ( int ind = 0; ind < 7; ind++ )
    {
        cin >> arr [ ind ];
    }

    cout << endl;

    for ( int ind = 0; ind < 7; ind++ )
```

```
{  
    if ( arr [ ind ] % 5 == 0 )  
    {  
        multiple ++;  
    }  
}
```

```
cout << "There are " << multiple << " multiples of  
five in the array." << endl;
```

```
return 0;  
}
```

Output

```
PS E:\Lab Tasks\Lab Task 7> g++ lab-task7.2.cpp -o lab-task7.2.exe  
PS E:\Lab Tasks\Lab Task 7> ./lab-task7.2.exe  
Enter seven integers:  
0  
2  
5  
99  
55  
7  
1005  
  
There are 4 multiples of five in the array.
```

Task – 3

```
#include<iostream>

using namespace std;

int main()
{
    int arr [ 10 ];

    cout << "Enter ten integers: " << endl;

    for ( int ind = 0; ind < 10; ind++ )
    {
        cin >> arr [ ind ];
    }

    cout << endl;

    cout << "Array in reverse order: " << endl;

    for ( int ind = 0; ind < 10; ind++ )
```



```
{  
    cout << arr [ 9 - ind ] << " ";  
}  
  
return 0;  
}
```

Output

```
PS E:\Lab Tasks\Lab Task 7> g++ lab-task7.3.cpp -o lab-task7.3.exe  
PS E:\Lab Tasks\Lab Task 7> ./lab-task7.3.exe  
Enter ten integers:  
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
  
Array in reverse order:  
10 9 8 7 6 5 4 3 2 1
```

Task – 4

```
#include<iostream>

using namespace std;

int positive_numbers ( int arr [ ] )
{
    int p = 0;

    for ( int ind = 0; ind < 10; ind++ )
    {
        if ( arr [ ind ] > 0 )
        {
            p++;
        }
    }

    return p;
}
```

```
int negative_numbers ( int arr [ ] )  
{  
    int n = 0;  
  
    for ( int ind = 0; ind < 10; ind++ )  
    {  
        if ( arr [ ind ] < 0 )  
        {  
            n++;  
        }  
    }  
  
    return n;  
}
```

```
int odd_numbers ( int arr [ ] )
```

```
{  
    int o = 0;  
  
    for ( int ind = 0; ind < 10; ind++ )  
    {  
        if ( arr [ ind ] % 2 != 0 )  
        {  
            o++;  
        }  
    }  
  
    return o;  
}
```

```
int even_numbers ( int arr [ ] )  
{  
    int e = 0;
```

```
for ( int ind = 0; ind < 10; ind++ )  
{  
    if ( arr [ ind ] % 2 == 0 )  
    {  
        e++;  
    }  
}  
  
return e;  
}
```

```
int zero_numbers ( int arr [ ] )  
{  
    int z = 0;  
  
    for ( int ind = 0; ind < 10; ind++ )
```

```
{  
    if ( arr [ ind ] == 0 )  
    {  
        z++;  
    }  
}  
  
return z;  
}  
  
int main()  
{  
    int num [ 10 ];  
    cout << "Enter ten integers: " << endl;  
  
    for ( int ind = 0; ind < 10; ind++ )  
    {
```

```
    cin >> num [ ind ];  
}  
  
cout << endl;  
  
    cout << "There are " << positive_numbers ( num )  
<< " positive numbers." << endl;  
    cout << "There are " << negative_numbers ( num  
) << " negative numbers." << endl;  
    cout << "There are " << odd_numbers ( num ) <<  
" odd numbers." << endl;  
    cout << "There are " << even_numbers ( num ) <<  
" even numbers." << endl;  
    cout << "There are " << zero_numbers ( num ) <<  
" zero numbers." << endl;  
  
    return 0;  
}
```

Output

```
PS E:\Lab Tasks\Lab Task 7> g++ lab-task7.4.cpp -o lab-task7.4.exe
PS E:\Lab Tasks\Lab Task 7> ./lab-task7.4.exe
Enter ten integers:
-1
34
97
43
0
23
-4532
90
0
87

There are 6 positive numbers.
There are 2 negative numbers.
There are 5 odd numbers.
There are 5 even numbers.
There are 2 zero numbers.
```

THE END